

# Intern Task Brief

## Hydropower Website Audit, Tech-Stack Analysis & Redesign Concepts

|                         |                                                                                                                                                 |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
|                         |                                                                                                                                                 |
| <b>Task</b>             | Audit 50+ hydropower company websites, analyze their tech stacks, identify how they could be better, and produce professional redesign concepts |
| <b>Assigned by</b>      | CI Tech                                                                                                                                         |
| <b>Assigned to</b>      | Krishna Gharti                                                                                                                                  |
| <b>Estimated effort</b> | ~3 weeks                                                                                                                                        |
| <b>Status</b>           | Assigned                                                                                                                                        |

---

### 1. Context (why this matters)

We're building a **premium, productized website platform for hydropower companies in Nepal**. Before we design and build, we need a rigorous, evidence-based picture of the current landscape: what these companies' websites actually look like, what technology they run on, and — most importantly — where they fall short. Your audit becomes the factual foundation for our product's design and feature decisions.

Do this well and your findings will directly shape a real product. Treat it as research a product team will make decisions from, not a box-ticking exercise.

---

### 2. Objective

Compile and analyze **at least 50 hydropower company websites** (Nepal-focused, plus a small set of international best-in-class benchmarks). For each, record its tech stack, evaluate it against a fixed scoring rubric, and document concrete weaknesses. Then synthesize the

findings and produce **professional redesign concepts** for a shortlist of the highest-priority sites.

---

### 3. Deliverables

You will produce **five** things:

| #  | Deliverable                                                                                                                                                  | Format                                           |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| D1 | <b>Website audit tracker</b> — 50+ sites, one row each, all columns filled (see §6 for schema)                                                               | Google Sheet / Excel                             |
| D2 | <b>Tech-stack analysis</b> — summary of what technologies dominate the sector and why it matters                                                             | Section in the report + a column in D1           |
| D3 | <b>Gap analysis</b> — the recurring weaknesses across the sector, ranked by how common and how serious they are                                              | Section in the report                            |
| D4 | <b>Redesign prototypes</b> — professional, working <b>React</b> builds for the top 2–3 priority sites (homepage + at least one inner page, fully responsive) | React code (repo + running prototype)            |
| D5 | <b>Findings report / deck</b> — a clear summary of everything above with your recommendations                                                                | Slides or a short doc (10–15 slides / 4–6 pages) |

---

### 4. Detailed instructions

#### Part A — Build the list (≥ 50 sites)

1. Start from the **seed list in §9** (verified companies + where to find more).
2. Expand it using the authoritative sources in §9 until you have **at least 50 sites with a working official website**. There are ~48+ hydropower companies listed on NEPSE alone, plus many unlisted ones, so hitting 50 is very achievable.
3. **Verify each URL is the company's official site** — not an aggregator (merolagani, sharesansar), not a news article, not a NEPSE profile page. If a company has no working site, note that (it's a finding in itself) and don't count it toward the 50.

4. Include **3–5 international best-in-class benchmarks** (see §9) so we have a "north star" for what professional design looks like. Mark these clearly as benchmarks, not competitors.

## Part B — Detect the tech stack

For each site, identify the technology it runs on. Use these tools/methods (record how you found it):

- **Wappalyzer** (browser extension) — CMS, frameworks, analytics, CDN.
- **BuiltWith.com** — deeper stack + hosting.
- **View source / DevTools** → **Network** — look at HTML structure, JS bundles, meta generator tags.
- **Response headers** — *Server*, *X-Powered-By*, *Set-Cookie* patterns often reveal the platform.
- **Tell-tale paths** — e.g. */wp-content/* → WordPress; Joomla *meta generator* tag; Laravel */storage/* paths + Cloudflare email-protection; *\_next/* → Next.js.
- **PageSpeed Insights / Lighthouse** — for the performance, accessibility, SEO, and best-practices scores (record the numbers).

Capture at minimum: CMS/framework, hosting/CDN, whether it's mobile-responsive, whether it uses HTTPS, and any analytics.

## Part C — Score each site (rubric in §7)

Evaluate every site against the **fixed rubric** on a 1–5 scale per criterion. Be consistent and objective — the same standard applies to every site. Add a one-line justification for any score of 1 or 5 so the scoring is defensible.

## Part D — Gap analysis ("how they could be better")

Don't just list scores. For each site, note its **top 2–3 concrete weaknesses** (e.g. "no mobile menu — breaks below 768px", "hero is a 3MB unoptimized JPG", "no live share price", "content only in English"). Then, in the report, **aggregate** these into sector-wide patterns: which problems show up most often, and which are most damaging. Rank them.

## Part E — Professional redesign (build it in React)

Pick the **2–3 highest-priority sites** (a mix of big incumbents with dated sites and important companies with weak sites). For each, build a redesign as a **working React prototype** — not static mockups:

- **Homepage + at least one inner page** (e.g. a Project detail or Investor Relations page).

- **Fully responsive** and mobile-first — it must work from 320px up, tested in a real browser, not just a design frame.
- **Real motion implemented** (see §8): hero parallax/reveal, scroll-reveal sections, animated stat counters, card hover states — actually running, not annotated.
- Apply a coherent, premium design direction (see §8) consistent with our product's system.
- Use realistic placeholder content (real company facts where easy), not lorem ipsum, so the layout is judged against real data density.
- Include a short **rationale** (in the repo README): what was wrong with the original, what you changed, and why it's better.

**Recommended stack:** Vite + React + Tailwind CSS, with Framer Motion (or Motion One) for animation. Next.js is fine if you prefer. Component-based structure, clean and readable code. Provide a running prototype (local run instructions in the README, or a deployed preview link via Vercel/Netlify).

You're demonstrating *how these sites should look and feel* as a real interface — layout, hierarchy, typography, color, imagery treatment, responsiveness, and live motion. It doesn't need backend/CMS wiring; front-end prototype quality is what's assessed.

---

## 5. Tools you'll use

Wappalyzer · BuiltWith · Chrome DevTools · Google PageSpeed Insights / Lighthouse · **React + Vite + Tailwind CSS + Framer Motion** (redesign build) · Google Sheets/Excel (tracker) · optionally a screenshot tool for evidence. All free.

---

## 6. Audit tracker schema (D1 columns)

Every site = one row. Fill **every** column.

| Column           | What goes in it                           |
|------------------|-------------------------------------------|
| Company name     | Official name                             |
| Website URL      | Verified official URL                     |
| Site works?      | Yes / No / Partial (broken links, errors) |
| Listed on NEPSE? | Yes / No + symbol if known                |

| Column                          | What goes in it                                            |
|---------------------------------|------------------------------------------------------------|
| Company size/type               | Large incumbent / mid / small / holding group / govt       |
| CMS / framework                 | e.g. Joomla, WordPress, Laravel+Bootstrap, Next.js, custom |
| Hosting / CDN                   | e.g. shared PHP host, Cloudflare, Vercel                   |
| Responsive?                     | Yes / Partial / No (test at 375px)                         |
| HTTPS?                          | Yes / No                                                   |
| Bilingual (EN/NE)?              | Full / Partial / English-only / Nepali-only                |
| Has live share price?           | Yes / No                                                   |
| Has interactive map?            | Yes / No / Static only                                     |
| Has investor-relations hub?     | Yes / No / Scattered PDFs                                  |
| Any animation/motion?           | None / Basic / Good                                        |
| Lighthouse Performance (mobile) | 0–100                                                      |
| Lighthouse Accessibility        | 0–100                                                      |
| Lighthouse SEO                  | 0–100                                                      |
| Design score                    | 1–5 (rubric)                                               |
| UX/navigation score             | 1–5                                                        |
| Responsiveness score            | 1–5                                                        |
| Content completeness score      | 1–5                                                        |
| Performance score               | 1–5                                                        |
| <b>Overall score</b>            | Average of the 5 rubric scores                             |
| Top weaknesses                  | 2–3 concrete issues                                        |
| Screenshot link                 | Homepage screenshot (desktop + mobile)                     |
| Notes                           | Anything else notable                                      |

---

## 7. Scoring rubric (1–5 per criterion)

Apply the **same** standard to every site.

| Criterion                      | 1 (poor)                              | 3 (average)                 | 5 (excellent)                                              |
|--------------------------------|---------------------------------------|-----------------------------|------------------------------------------------------------|
| <b>Design / visual</b>         | Dated template, cramped, inconsistent | Clean but generic template  | Distinctive, premium, coherent brand system                |
| <b>UX / navigation</b>         | Confusing, deep menus, dead ends      | Usable, some friction       | Clear, shallow IA, everything findable fast                |
| <b>Responsiveness</b>          | Breaks on mobile                      | Works but cramped on mobile | True mobile-first, flawless to 320px                       |
| <b>Content completeness</b>    | Missing key sections                  | Has the basics              | Complete: IR, projects, notices, media, careers, bilingual |
| <b>Performance / technical</b> | Slow, heavy, no HTTPS/SEO             | Acceptable                  | Fast (LCP<2.5s), optimized, good SEO/a11y                  |

---

## 8. Design language to follow (for D4 redesigns)

Keep redesigns consistent with our product direction so they read as one coherent vision:

- **Mood:** editorial, spacious, cinematic — grounded in water, energy, and the Himalaya. Not another blue Bootstrap template.
- **Color (default):** deep hydro navy #0A2540 base, glacier teal #14B8A6 accent, current blue #1E6FEB secondary, warm sunrise #F5A623 for energy/stat highlights, clean neutrals, and a proper dark mode.
- **Typography:** modern, legible sans for UI/body (e.g. Inter); optional refined display face for heroes; a well-matched Devanagari face (e.g. Noto Sans Devanagari / Mukta) sized to match the Latin type.

- **Imagery:** full-bleed project/river/turbine photography with gradient/duotone overlays for cohesion and text contrast; prefer real project photos over stock.
  - **Motion (build it in React):** hero parallax/reveal, scroll-reveal on sections, animated stat counters (MW / GWh / CO<sub>2</sub>), project-card hover lift, animated charts. Implement with Framer Motion / Motion One. Refined, never busy; must respect `prefers-reduced-motion`.
- 

## 9. Seed list & where to find more

### Verified official sites (start here)

- Butwal Power Company — <https://www.bpc.com.np/> (*Joomla — good example of a dated incumbent*)
- Chilime Hydropower — <https://www.chilime.com.np/en/>
- National Hydro Power Company — <https://www.nhpcl.com/>
- Nepal Water & Energy Development Co. (NWEDC) — <http://nwedcpl.com/>
- Dovan Hydropower — <https://www.dovanhydropower.com.np/> (*Laravel+Bootstrap — newest of the set; good SEO, generic look*)
- Peoples Hydropower — <https://peopleshydro.com.np/>
- Himalayan Hydro Power — <https://hhpl.com.np/>

### Companies to find official sites for (all real, mostly NEPSE-listed)

Upper Tamakoshi Hydropower · Rasuwagadhi Hydropower · Sahas Urja · Api Power Company · Arun Valley Hydropower · Sanima Mai Hydropower · Mountain Energy Nepal (Mistri Khola) · Nepal Hydro Developer · Supermai Hydropower · Green Ventures (Likhu-IV) · Ru Ru Jalbidhyut Pariyojana · Mandu Hydropower · Menchhiyam Hydropower · Dibyashwori Hydropower · Mountain Hydro Nepal · Sanigad Hydro · Yambaling Hydropower ... and the rest of the NEPSE hydropower sector.

### Authoritative rosters to reach 50+

- **NEPSE** company list, Hydropower sector — <https://www.nepalstock.com/>
- **Merolagani** company list — <https://merolagani.com/CompanyList.aspx> (*directory only — then find each company's own site*)
- **IPPAN** (Independent Power Producers' Association Nepal) member directory — <https://www.ippan.org.np/>
- **Department of Electricity Development** licensee lists — <https://doed.gov.np/>
- Sharesansar hydropower listings (*directory only*)

International best-in-class benchmarks (3–5, mark as benchmarks)

Statkraft · Hydro-Québec · SN Power · Voith Hydro · Andritz Hydro — study these for what "professional, premium energy-company web design" looks like.

---

## 10. Timeline & milestones

| Week          | Focus                                                                  | Milestone                                       |
|---------------|------------------------------------------------------------------------|-------------------------------------------------|
| <b>Week 1</b> | Build the list (50+), start tech-stack + Lighthouse data               | Tracker populated with all sites + tech columns |
| <b>Week 2</b> | Complete scoring, screenshots, per-site weaknesses; write gap analysis | D1 fully filled + D3 draft                      |
| <b>Week 3</b> | Build 2–3 redesigns in React + final report/deck                       | D4 + D5 delivered                               |

Check in with me at the end of each week.

---

## 11. Definition of Done

- ☐ **50+ verified official sites** in the tracker, every column filled.
  - ☐ Tech stack identified for each (with method noted) + Lighthouse scores recorded.
  - ☐ Every site scored on all 5 rubric criteria, consistently.
  - ☐ Sector-wide gap analysis with weaknesses **ranked** by frequency and severity.
  - ☐ **2–3 redesign prototypes** built in **React** (homepage + inner page, fully responsive, real motion) with rationale in the README.
  - ☐ Findings report/deck summarizing everything with clear recommendations.
- 

## 12. Tips & guardrails

- **Be objective and consistent** — same rubric standard for every site. Justify every 1 and 5.



- **Evidence over opinion** — record scores, screenshots, and Lighthouse numbers, not just impressions.
- **Verify official sites** — don't audit aggregators or news pages by mistake.
- **Test mobile for real** — use DevTools device mode at 375px and 320px, not just a narrowed desktop window.
- **Don't scrape aggressively** — you're browsing and auditing normally, not hammering servers or downloading whole sites.
- **Cite your sources** — note where you found each URL and each tech-stack finding.
- **Flag surprises** — if you find a genuinely good Nepali hydropower site, that's a valuable finding; call it out.
- **Ask early** if the list, rubric, or design expectations are unclear — don't guess for three weeks.

---

*Questions? Message me. Have fun with it — the redesigns especially are your chance to show what you can do.*